



16 July 2019

2DEG Correlation Energy

An Approach for the Correlation energy calculation
of two-dimensional electron gas systems

Nivedita



Table and Block

TableTitle	
CapA	lots of examplesA
CapB	lots of examplesB
CapC	lots of examplesC

BlockTitle

something to emphasise



Two Figures aside

fooA.pdf

fooB.pdf

Figure 1: Caption of Figure A

Figure 2: Caption of Figure B



Text and Figure aside

fooC.pdf

Some text and a bullet point list

- ItemA
- ItemB
- ItemC
- ItemD

Figure 3: Caption of Figure C



One Figure

fooD.pdf

Figure 4: Caption of Figure D



Adiabatic Connection



RPA

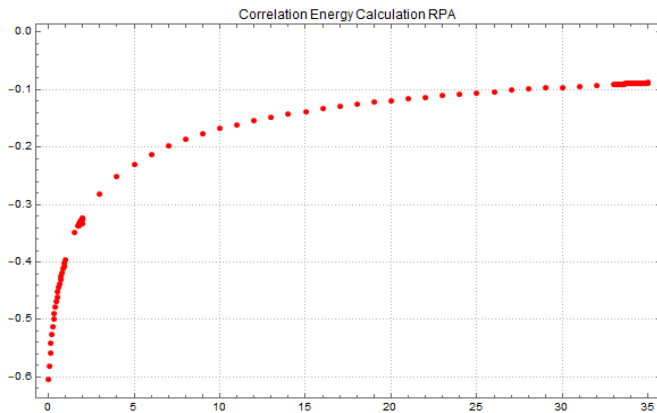
$$\frac{\epsilon_C^{\text{RPA}}}{Ry} = -\frac{2}{\pi r_s^2} \int_0^\infty dy y^2 \int_0^\infty d\nu \left\{ \frac{\sqrt{2} r_s}{y} f(y, \nu) - \log \left(1 + \frac{\sqrt{2} r_s}{y} f(y, \nu) \right) \right\}$$

$$f(y, \nu) = \frac{2 \operatorname{Re} \Psi \left(\frac{y}{2} + i\nu \right)}{y}$$

$$\Psi(z) = \int_0^1 dx x \int \frac{d\theta}{2\pi} \frac{1}{z - x \cos(\theta)}$$

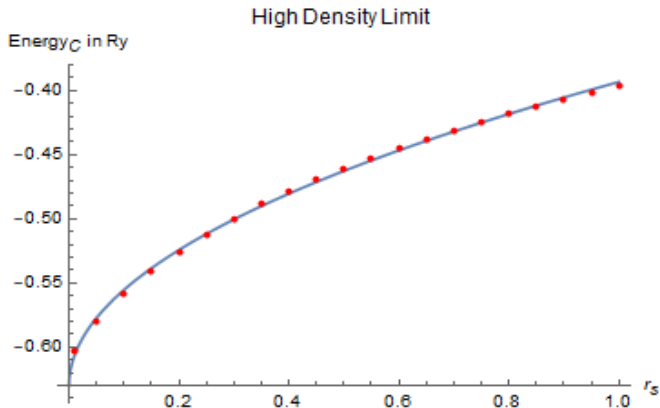


Plot





Plot





Plot

